

General Relativity Notes: Tensors & Galilean Limit

Part 1: The Einstein Field Equations & Galileo's Falling Stone

The core equation of General Relativity—the **Einstein Field Equation (EFE)**—is an equality between symmetric rank-2 tensors in 4-dimensional spacetime:

$$G_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu} \quad (1)$$

where $G_{\mu\nu}$ (the Einstein tensor) encodes spacetime curvature, and $T_{\mu\nu}$ (the stress-energy-momentum tensor) encodes matter and energy.

The metric tensor $g_{\mu\nu}$ lives implicitly inside $G_{\mu\nu}$, built from its derivatives:

$$G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu} \quad (2)$$

1. The Background Spacetime (The Earth)

Outside the Earth, where the stone falls, local space is effectively empty of ambient mass. Thus, $T_{\mu\nu} \approx 0$. The field equation simplifies to $G_{\mu\nu} = 0$, yielding the **Weak-Field Schwarzschild Metric** in $(c dt, dr, d\theta, d\phi)$ coordinates:

$$g_{\mu\nu} \approx \begin{pmatrix} -\left(1 - \frac{2GM}{c^2 r}\right) & 0 & 0 & 0 \\ 0 & \left(1 - \frac{2GM}{c^2 r}\right)^{-1} & 0 & 0 \\ 0 & 0 & r^2 & 0 \\ 0 & 0 & 0 & r^2 \sin^2 \theta \end{pmatrix} \quad (3)$$

The g_{00} component contains the Newtonian gravitational potential $\Phi(r) = -GM/r$, providing the uniform downward acceleration $g \approx 9.81 \text{ m/s}^2$.

2. The Stress-Energy Tensor of the Falling Stone

Treating the falling stone of mass m as a localized dust particle with 4-velocity $u^\mu = \frac{dx^\mu}{d\tau}$, its stress-energy tensor is:

$$T_{\text{stone}}^{\mu\nu}(x) = m u^\mu u^\nu \frac{\delta^{(3)}(\vec{x} - \vec{x}_{\text{stone}}(\tau))}{\gamma} \quad (4)$$

For non-relativistic velocities $v \ll c$, where $u^\mu \approx (c, v_r, 0, 0)$:

$$T_{\text{stone}}^{\mu\nu} \approx \begin{pmatrix} \rho c^2 & \rho c v_r & 0 & 0 \\ \rho c v_r & \rho v_r^2 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \quad (5)$$

- ρc^2 (T^{00}): Rest-mass energy density of the stone.
- $\rho c v_r$ (T^{01}): Radial momentum density as it falls.
- ρv_r^2 (T^{11}): Radial momentum flux (negligible compared to ρc^2).

3. Connecting the Motion

Since $m \ll M$, the stone acts as a test particle following a **geodesic** in Earth's metric:

$$\frac{d^2 x^\mu}{d\tau^2} + \Gamma_{\alpha\beta}^\mu \frac{dx^\alpha}{d\tau} \frac{dx^\beta}{d\tau} = 0 \quad (6)$$

Taking the spatial radial component $\mu = r$ in the weak-field limit:

$$\frac{d^2 r}{dt^2} \approx -\frac{1}{2}c^2 \partial_r g_{00} = -\frac{GM}{r^2} = -g \quad (7)$$

Part 2: Step-by-Step Derivation of $d^2r/dt^2 = -g$

1. The Weak-Field Static Metric

In the weak-field limit ($GM/c^2r \ll 1$), using coordinates $(x^0, x^1, x^2, x^3) = (ct, r, \theta, \phi)$ and potential $\Phi(r) = -\frac{GM}{r}$:

$$g_{\mu\nu} \approx \begin{pmatrix} -\left(1 + \frac{2\Phi}{c^2}\right) & 0 & 0 & 0 \\ 0 & 1 - \frac{2\Phi}{c^2} & 0 & 0 \\ 0 & 0 & r^2 & 0 \\ 0 & 0 & 0 & r^2 \sin^2 \theta \end{pmatrix} \quad (8)$$

The relevant metric components and their inverses are:

$$g_{00} \approx -\left(1 + \frac{2\Phi}{c^2}\right), \quad g_{rr} \approx 1 - \frac{2\Phi}{c^2} \quad (9)$$

$$g^{00} \approx -\left(1 - \frac{2\Phi}{c^2}\right), \quad g^{rr} \approx 1 + \frac{2\Phi}{c^2} \quad (10)$$

2. Deriving the Christoffel Symbol Γ_{00}^r

The Christoffel symbols of the second kind are defined as:

$$\Gamma_{\alpha\beta}^\lambda = \frac{1}{2}g^{\lambda\sigma} (\partial_\alpha g_{\beta\sigma} + \partial_\beta g_{\alpha\sigma} - \partial_\sigma g_{\alpha\beta}) \quad (11)$$

For purely radial motion with $v \ll c$, the temporal coordinate derivative $dx^0/d\tau = c dt/d\tau$ dominates. The required radial symbol is:

$$\Gamma_{00}^r = \frac{1}{2}g^{rr} (\partial_0 g_{0r} + \partial_0 g_{r0} - \partial_r g_{00}) \quad (12)$$

Since the metric is static ($\partial_0 g_{\alpha\beta} = 0$):

$$\Gamma_{00}^r = -\frac{1}{2}g^{rr} \partial_r g_{00} \quad (13)$$

Substituting $g^{rr} \approx 1$ and $g_{00} = -\left(1 + \frac{2\Phi}{c^2}\right)$:

$$\Gamma_{00}^r \approx -\frac{1}{2}(1) \frac{\partial}{\partial r} \left(-1 - \frac{2\Phi}{c^2}\right) = \frac{1}{c^2} \frac{d\Phi}{dr} \quad (14)$$

With $\frac{d\Phi}{dr} = \frac{GM}{r^2}$:

$$\Gamma_{00}^r \approx \frac{GM}{c^2 r^2} \quad (15)$$

3. The Geodesic Equation

The radial component ($\mu = r$) of the geodesic equation is:

$$\frac{d^2 r}{d\tau^2} + \Gamma_{00}^r \left(\frac{dx^0}{d\tau}\right)^2 + 2\Gamma_{0r}^r \left(\frac{dx^0}{d\tau}\right) \left(\frac{dr}{d\tau}\right) + \Gamma_{rr}^r \left(\frac{dr}{d\tau}\right)^2 = 0 \quad (16)$$

In the non-relativistic limit ($v \ll c$):

- $\frac{dx^0}{d\tau} = c \frac{dt}{d\tau} \approx c$ (since $dt/d\tau \approx 1$).
- Terms containing $\frac{dr}{d\tau}$ are suppressed by factors of v/c and v^2/c^2 .

Retaining leading order terms:

$$\frac{d^2 r}{d\tau^2} + \Gamma_{00}^r \left(c \frac{dt}{d\tau}\right)^2 \approx 0 \implies \frac{d^2 r}{dt^2} + c^2 \Gamma_{00}^r \approx 0 \quad (17)$$

4. Final Result

Substituting Γ_{00}^r :

$$\frac{d^2 r}{dt^2} + c^2 \left(\frac{GM}{c^2 r^2} \right) = 0 \implies \frac{d^2 r}{dt^2} = -\frac{GM}{r^2} \quad (18)$$

Evaluating at Earth's surface ($r \approx R_{\oplus}$), where $g \equiv \frac{GM}{R_{\oplus}^2} \approx 9.81 \text{ m/s}^2$:

$$\frac{d^2 r}{dt^2} = -g \quad (19)$$